# Week 7: Auctions I

## Bidding, Information, and Auction Rules

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Semester 2026–27 Term 1

When values are private, auction rules shape what bidders reveal and how aggressively they compete.

# From Week 6 to Week 7

## Week 6: Repeated Games

- Future rewards and punishments
- History-dependent strategies
- Discount factors and patience
- Cooperation over time

## Week 7: Auctions

- Private information
- Allocation and payment rules
- Strategic bidding
- Mechanism choice

**Repetition changes incentives across time; auctions change incentives through rules.**

# Opening Decision: A Licence Worth HK$100,000

## The Situation

You and a rival submit sealed bids for one licence. The highest bid wins. You value the licence at **HK$100,000**, but you do not know the rival's value or bid.

## Rule A: First Price

- Highest bid wins.
- Winner pays own bid.

## Rule B: Second Price

- Highest bid wins.
- Winner pays the highest losing bid.

**Would you submit the same bid under both rules?**

# Today’s Learning Outcomes

## By the end of class, you should be able to:

1. define an auction as an allocation-and-payment mechanism;
2. distinguish first-price, second-price, English, and Dutch auctions;
3. distinguish private-value and common-value settings;
4. explain truthful bidding in a second-price auction;
5. explain bid shading in a first-price auction;
6. interpret revenue equivalence and the winner’s curse;
7. apply auction design to platforms, spectrum, and procurement.

# What Is an Auction?

## Definition

An auction is a mechanism that maps participants' bids or messages into an **allocation** and a **payment**.

**The allocation problem**

- One seller may face many buyers.
- The object or right is scarce.
- Individual bargaining may be impractical.

**The information problem**

- The seller does not know each buyer's value.
- Each bidder knows more about its own value.
- Rules must elicit enough information to allocate the object.

**Who receives the object, and how much does the winner pay?**

# Why Private Information Changes the Game

- A bidder knows its own valuation, cost, or budget.
- Rivals' valuations are not directly observed.
- If every valuation were known, bidding would often become straightforward.
- With uncertainty, a bidder must reason about both rivals and the auction rule.

## Strategic Question

Do not ask only, "What is the object worth to me?" Also ask, "How does the mechanism use my bid?"

# Private Value Versus Common Value

## Private Value

A bidder's value depends primarily on its own preferences or use.

- Collectibles
- Personal-use items
- Art valued for enjoyment

## Common Value

The underlying asset value is broadly shared, but bidders have different estimates or signals.

- Extraction rights
- Resale opportunities
- Some spectrum licences

**Would learning rivals' information change your estimate of the object's value?**

# First-Price Versus Second-Price Sealed Bids

## First-Price Sealed Bid

- Each bidder submits one confidential bid.
- The highest bid wins.
- The winner pays its own bid.
- Typical incentive: bid below value.

## Second-Price Sealed Bid

- Each bidder submits one confidential bid.
- The highest bid wins.
- The winner pays the highest losing bid.
- Standard incentive: bid true value.

## Key Difference

The allocation rule is the same, but the payment rule changes the bidding incentive.

# Open Auctions: English and Dutch

| **Format** | Process | Standard Private-Value Logic |
|---|---|---|
| **English** | Price rises until only one bidder remains. | Resembles a second-price auction: the winner pays near the point at which the second-highest bidder exits. |
| **Dutch** | Price falls until one bidder accepts. | Resembles a first-price auction: the bidder chooses when to stop and pays that price. |

**English $\approx$ second price; Dutch $\approx$ first price.**

# Second-Price Auction: Bid Your Value

## Standard Result

In a private-value second-price auction, bidding your true valuation is a **weakly dominant strategy**.

## Highest Rival Bid Below Your Value

- You want to win.
- Bidding your value wins.
- You pay the rival's lower bid.
- Your surplus is positive.

## Highest Rival Bid Above Your Value

- You want to lose.
- Bidding your value loses.
- Winning would require paying too much.
- Truthful bidding avoids the loss.

# Worked Check: Underbidding

## Your Value Is 100; Highest Rival Bid Is 80

- Bid 100: you win, pay 80, and receive surplus $100 - 80 = 20$.
- Bid 70: you lose even though buying at 80 would have been profitable.

## Why Underbidding Can Hurt

A lower bid does not reduce the second-price payment when you win. It can only cause you to lose an outcome you wanted.

# Worked Check: Overbidding

## Your Value Is 100; Highest Rival Bid Is 110

- Bid 100: you lose and pay nothing.
- Bid above 110: you may win and pay 110, producing surplus $100 - 110 = -10$.

## Why Overbidding Can Hurt

A bid above value can make you win precisely when you would prefer to lose.

**Your bid selects the winning cutoff; a rival's bid determines the payment.**

# First-Price Auction: The Bid-Shading Trade-off

## Bid Higher

- Greater probability of winning
- Useful against aggressive rivals
- Lower surplus if you win

## Bid Lower

- Greater surplus if you win
- Useful against conservative rivals
- Greater probability of losing

## Bid Shading

A bidder normally submits less than its value because bidding the full value leaves no surplus when it wins.

# What Determines a First-Price Bid?

- Your valuation of the object
- The number of competitors
- Beliefs about rivals' valuations and bids
- Risk attitudes
- Reserve prices and minimum increments
- Information revealed during the auction

**Balance the probability of winning against the profit conditional on winning.**

# Revenue Equivalence: A Benchmark

## Benchmark Result

Under standard assumptions, common auction formats can generate the same expected seller revenue.

**Standard assumptions**

- Risk-neutral bidders
- Symmetric bidders
- Independent private values
- Highest-value bidder receives the object

**Why formats can differ**

- Risk aversion
- Bidder asymmetry
- Common or affiliated values
- Reserve prices or platform rules

# Revenue Equivalence: Simple Intuition

## Suppose Valuations Are 10, 9, 8, and 7

- In a second-price auction, truthful bids make the value-10 bidder win and pay 9.
- In a first-price auction, the highest-value bidder shades its bid because it pays its own bid.
- Under the benchmark model, equilibrium bid shading adjusts so that expected revenue matches the relevant competing valuation.

## Interpretation

The payment rules differ, but strategic adjustment can offset the apparent revenue difference on average.

# Winner's Curse

## Common-Value Logic

When bidders estimate the same underlying asset value, the winner is often the bidder with the most optimistic estimate.

- Winning is informative: every rival estimated less or bid less.
- The unconditional estimate may therefore be too optimistic conditional on winning.
- A rational bidder should adjust for the information contained in the event of winning.

## Managerial Warning

Winning is not success if the payment exceeds the asset's underlying value.

# Proxy Bidding on a Platform

## How It Works

A bidder enters a maximum willingness to pay. The platform raises the standing bid only as needed and stops at the stated limit.

1. Enter the maximum amount the item is worth to you.
2. The system raises your standing bid by the required increment.
3. If a rival remains below your maximum, you stay in front.
4. If a rival exceeds your maximum, the system stops bidding for you.

## Design Details

Bid increments, closing rules, reserve prices, and information disclosure can change behaviour.

# Digital Advertising Auctions

## The Allocation Problem

A platform has limited advertising positions. Multiple advertisers value a position differently and compete through bids.

**Bidder questions**

- What is a click or conversion worth?
- How much should the bid be shaded?
- How valuable is a particular keyword?

**Platform questions**

- First-price or second-price payment?
- How should relevance or quality enter ranking?
- What if bidders are highly asymmetric?

**A theoretically elegant rule may perform differently when competition is thin or bidders are asymmetric.**

# Spectrum Auctions: Multiple Related Licences

## Complementarity

- Neighbouring regions may be more valuable together.
- A consistent band may support network deployment.
- Winning one licence can raise another licence's value.

## Substitutability

- Some licences perform similar functions.
- Technical constraints may reduce joint value.
- Winning one licence can lower another licence's value.

## Design Challenge

Selling related licences is a package-design problem, not merely a sequence of independent single-item auctions.

# Reverse Auctions and Procurement

## Ordinary Auction

- One seller, many buyers
- Highest acceptable bid wins
- Used to sell an asset or right

## Reverse Auction

- One buyer, many suppliers
- Lowest acceptable offer wins
- Used for sourcing and procurement

## Reverse Second Price

The lowest-cost supplier wins but is paid the second-lowest offer.

# Procurement: Price Is Not the Only Dimension

## Price Score

- Lower price
- Announced formula
- Delivery risk

## Quality Score

- Technical capability
- Reliability
- Verifiable evidence

## Combined Award

- Stated weights
- Minimum standards
- Best evaluated offer

## Managerial Lesson

A low bid is valuable only if the supplier can deliver the required quality.

# Open Versus Sealed Bidding

## Open Bidding

- Participants observe the price path.
- Rival behaviour may reveal information.
- Useful when others' signals affect value estimates.
- May intensify rivalry or anchoring.

## Sealed Bidding

- Bids are submitted privately.
- Less information is revealed during bidding.
- Natural when values are largely independent.
- Requires beliefs about rivals before submission.

**The information rule can matter as much as the payment rule.**

# Application for Next Class: Digital Ad Opportunity

## The Situation

A digital advertising opportunity is worth **HK$50 per click** to your company.

1. What would you bid in a second-price auction, and why?
2. Why can bidding HK$30 lose a profitable opportunity?
3. Why can bidding HK$70 create a loss?
4. What would you bid in a first-price auction?
5. Which platform rules or market conditions change your answer?

# Checklist for an Auction Analysis

## Define the Environment

- What is being allocated?
- Who are the bidders?
- Private value or common value?
- Single item or related package?
- What information is observed?

## Solve and Interpret

- Who wins?
- What does the winner pay?
- Is truthful bidding optimal?
- What assumptions drive revenue?
- What rule changes behaviour?

## Final Check

Can you explain both the bidder's strategy and the seller's design choice?

# Application Debrief

1. Which part of the rule determines allocation?
2. Which part determines payment?
3. Does the bid affect only winning, or also the amount paid?
4. What must the bidder believe about competitors?
5. Which assumption could reverse the recommendation?

## Managerial Lesson

Auction performance depends on incentives, information, competition, and implementation details.

# Week 7: Main Takeaways

1. An auction maps bids into allocation and payment.
2. Private information makes the auction rule strategically important.
3. In a second-price auction, truthful bidding is weakly dominant under the standard model.
4. In a first-price auction, bidders trade off winning probability against surplus.
5. English auctions resemble second-price auctions; Dutch auctions resemble first-price auctions.
6. Common-value settings create winner's-curse concerns.
7. Revenue equivalence is a benchmark whose assumptions must be checked.
8. Real mechanisms add reserve prices, increments, quality rules, and information policies.

**Identify the value environment. Read the payment rule. Derive the incentive. Test the assumptions.**

# Exit Ticket and Looking Ahead

## Exit Ticket

Why is bidding true value sensible in a second-price auction? Why does the same reasoning fail in a first-price auction?

## Looking Ahead

Compare auction formats from the seller's perspective and examine how bidder asymmetry, risk, and platform rules change the benchmark predictions.

# Supplementary: Second-Price Dominance Check

Let your value be $v$, your bid be $b$, and the highest rival bid be $r$.

**If $r < v$**

You want to win and receive payoff $v - r > 0$. Bidding $v$ wins. An underbid below $r$ loses this surplus.

**If $r > v$**

You want to lose. Bidding $v$ loses. An overbid above $r$ wins and produces payoff $v - r < 0$.

**Conclusion**

Bidding $b = v$ is at least as good as every alternative bid for every possible highest rival bid.

# Supplementary: Common Auction Errors

| **Common Error** | Better Check |
|---|---|
| **Call the auction "second price" because there are two bidders.** | Identify the payment rule: does the winner pay the highest losing bid? |
| **Bid true value in every auction.** | Truthful bidding is the standard result for second price, not first price. |
| **Assume winning is always good.** | Compare the payment with value and consider the winner's curse. |
| **Apply revenue equivalence without qualifications.** | Check risk neutrality, symmetry, independence, and allocation assumptions. |
| **Evaluate procurement on price alone.** | State minimum quality and the price–quality scoring rule. |

# Supplementary: Auction-Rule Design Checklist

1. Define the eligible bidders and object or package being allocated.
2. State whether bids are open or sealed.
3. State the allocation rule and tie-breaking rule.
4. State the payment rule.
5. Decide whether to use a reserve price and minimum increment.
6. Specify the closing or activity rule.
7. Explain what information is disclosed before, during, and after bidding.
8. Identify quality, reliability, or non-price requirements.